5 **(a)** **(i)** Fig. 5.1 shows the cross section of a wire XY which carrying a constant current flowing out of the plane of the paper at right angles. Sketch the magnetic flux pattern due to the current. [2]

Fig. 5.1

(ii) The current-carrying wire XY in **(a)(i)** is placed in the region between the poles of a strong magnet. Sketch in Fig. 5.2 the resultant magnetic flux pattern in the region between the poles of a magnet. [2]

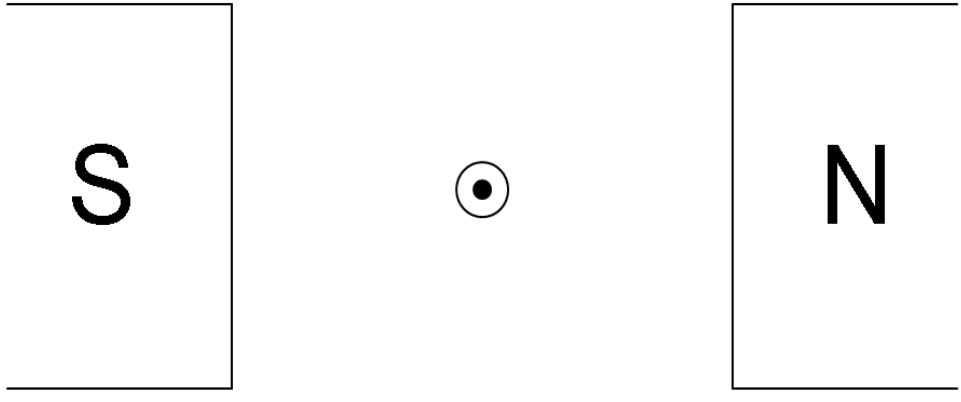

Fig. 5.2

(b) The setup in **(a)** is now placed on a top-pan balance as shown in Fig. 5.3 and Fig. 5.2 shows the front view. Outside the region between the poles of the strong magnet, the flux density can be considered as zero. When there is no current passing through XY, the top-pan balance is tared.

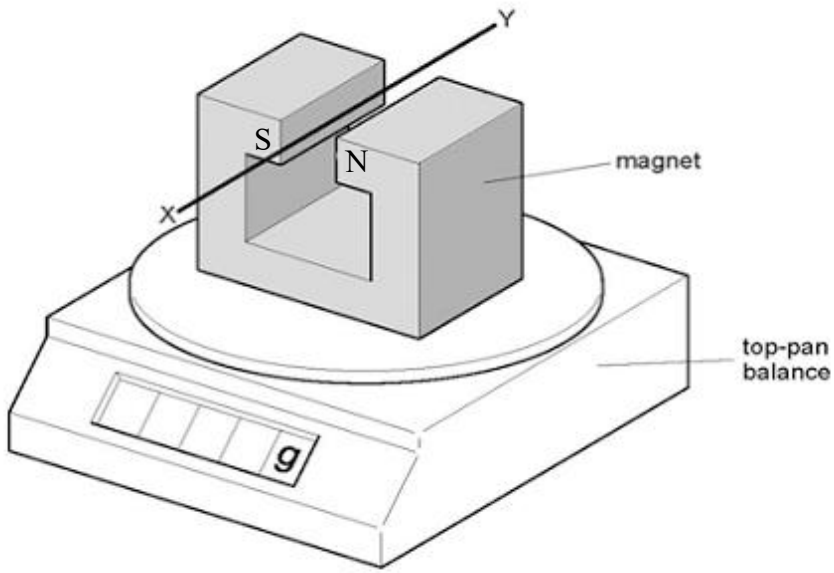

Fig. 5.3

A direct current is now passed through the wire in the direction from Y to X.

State and explain the observation in the reading on the top-pan balance.

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